

NATTAPAT (VERN) PRAYOONTHONG

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Education

Carnegie Mellon University, Carnegie Institute of Technology
B.S. in Mechanical Engineering, Minor in Robotics (3.45 GPA)

Pittsburgh, PA
Expected May 2028

Relevant Experience

CMU Lunabotics

Pittsburgh, PA

Mobility Subteam Lead

August 2025 - Present

- Led a 6-member mobility subteam developing a tracked drive system for an 80kg lunar rover
- **Coordinated** the transition of 20+ CAD components from Onshape to SolidWorks across a 6-member mobility team.
- **Designed and integrated** a 6-tooth, 5.4-in-diameter drive sprocket and 9 mechanical interfaces through 4 iterative redesigns, incorporating increasing DFM considerations to improve manufacturability, assembly, and fit
- **Manufactured** 60+ aluminum tread plates through precision cutting and CNC drilling, maintaining ± 0.01 in tolerance
- **Fabricated and integrated** 10+ track components using 3D printing and manual milling for the final rover assembly

National Science and Technology Development Agency (NSTDA)

Bangkok, Thailand

Intern

June 2025 - July 2025

- **Developed and tested** an automated object-counting system using image thresholding and contour detection
- **Curated and annotated** a custom dataset for training an oriented object-detection model
- **Trained and evaluated** a YOLOv11-OBb model to detect and count objects across varied orientations
- **Analyzed** performance across lighting and orientation conditions to identify failure cases and improve system robustness

Projects

Hammock Chair Design

Pittsburgh, PA

March 2026 - May 2026

- **Collaborated** with three teammates to design, fabricate, and test a portable load-bearing chair and rope-support system
- **Performed** hand calculations to determine structural loads and support mechanical design decisions
- **Conducted** finite element analysis (FEA) to evaluate stress and deformation in load-bearing components under expected loads
- **Built and tested** a physical prototype to verify stability, fit, and load-bearing performance

General Robotics

Pittsburgh, PA

August 2025 - July 2026

- **Designed and fabricated** mechanical components for an Urban Search and Rescue robot using SolidWorks
- **Designed** a custom mold for casting polyurethane wheels, considering geometry, fit, and manufacturability
- **Built and tested** robot components through iterative prototyping and performance evaluation
- **Developed and tuned** PID control algorithms for a self-balancing robot through repeated testing and data analysis

Posture Your Focus

Pittsburgh, PA

April 2025 - May 2025

- **Developed** a real-time computer vision application using Python, OpenCV, and MediaPipe to monitor 4 user behaviors, including slouching, screen distance, phone use, and inactivity
- **Implemented** posture analysis for 2 ergonomic indicators (slouching and viewing distance) to provide real-time feedback during desk work
- **Built** distraction monitoring for 2 focus-related behaviors, detecting phone use and lack of interaction and triggering customizable alerts

Skills

CAD & Mechanical Design: SolidWorks, Onshape, ANSYS FEA, CFD, 3D Printing, CAD drawings

Manufacturing & Prototyping: CNC, Manual Milling, Mechanical Assembly, Prototyping, Design Verification, DFM, GD&T

Programming & Data Analysis: Python, C, MATLAB, Computer Vision

Languages: Thai (Native), English (Fluent), Mandarin (Intermediate, HSK 4)

Relevant Coursework: Fluid Mechanics, Thermodynamics, Heat Transfer, Robot Kinematics & Dynamics, Principles of Imperative Computation